



UNIVERSIDAD NACIONAL DE LOJA

INGENIERÍA EN SISTEMAS/CARRERA
COMPUTACIÓN

ANÁLISIS NUMÉRICO

Examen

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EXAMEN U1

Fecha: 22/05/2026

Evaluación

Nombre: Maria Chica Medina

Pregunta ①

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!} + R$$

$$f(x) = 5x^4 - 2x^3 \quad \left. \begin{array}{l} a=1 \\ x=3 \end{array} \right\}$$

$$f(3) = 5(3)^4 - 2(3)^3 = 5(81) - 2(27)$$

$$= 405 - 54 = 351$$

$$f(x) = 5x^4 - 2x^3 \quad \left. \begin{array}{l} a=1 \\ f(1) = 5(1)^4 - 2(1)^3 = 3 \\ f'(1) = 20 - 6 = 14 \\ f''(1) = 60 - 12 = 48 \\ f'''(1) = 120 - 42 = 78 \\ f^{(4)}(1) = 120 \end{array} \right\}$$

• 5 términos $\rightarrow$ serie de Taylor

$$f(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{(4)}(1)}{4!}(x-1)^4$$

$$f(x) = 3 + 14(x-1) + 24(x-1)^2 + 18(x-1)^3 + 5(x-1)^4$$

• $x=3$

$$x-1=2$$

$$f(3) = 3 + 14(2) + 24(2)^2 + 18(2)^3 + 5(2)^4$$

$$f(3) = 3 + 28 + 96 + 144 + 80$$

$$f(3) = 351$$

$$\left. \begin{array}{l} f(3) = 5(3)^4 - 2(3)^3 \\ = 5(81) - 2(27) \\ = 405 - 54 \\ = 351 \end{array} \right\}$$

- Error relativo porcentual

$$Er(\%) = \left| \frac{V_{\text{obto}} - V_{\text{a}}}{V_{\text{verado}}} \right| \cdot 100$$

$$Er(\%) = \left| \frac{351 - 351}{351} \right| \cdot 100 = 0\% //$$

Pregunta ②

$$f(x) = x^3 - 155$$

$$f'(x) = 3x^2$$

- Newton-Raphson

$$X_{n+1} = X_n - \frac{f(X_n)}{f'(X_n)}$$

$$X_0 = 15$$

- Iteración 1

$$X_1 = 15 - \frac{15^3 - 155}{3(15)^2}$$

$$X_1 = 15 - \frac{3220}{675}$$

$$X_1 = 10.22963$$

$$e_1 = \left| \frac{X_1 - X_0}{X_1} \right| \cdot 100 = 46.63\%$$

- Iteración 2

$$X_2 = 10.22963 - \frac{10.22963^3 - 155}{3(10.22963)^2}$$

$$X_2 = 7.43762$$

$$e_2 = \left| \frac{7.43762 - 10.22963}{7.43762} \right| \cdot 100 = 37.54\%$$

$$e_2 = 37.54\%$$

- Iteración 3

$$X_3 = 7.43762 - \frac{7.43762^3 - 155}{3(7.43762)^2}$$

$$X_3 = 5.63228$$

$$e_3 = \frac{5.63228 - 7.43762}{5.63228} \cdot 100$$

$\Rightarrow$

$$e_3 = 32.05\% //$$

Octave

PREGUNTA 1

```
octave:2> pregunta2
```

Iter	x_nuevo	Error aprox %
1	10.229630	46.6329%
2	7.313484	39.8735%
3	5.841622	25.1961%

```
octave:3> pregunta2
```

Iter	x_nuevo	Error aprox %
1	10.229630	46.6329%
2	7.313484	39.8735%
3	5.841622	25.1961%

```
octave:4> pregunta2
```

Iter	x_nuevo	Error aprox %
1	10.229630	46.6329%
2	7.313484	39.8735%
3	5.841622	25.1961%

PREGUNTA 2

```
octave:1> preguntal
```

Symbolic pkg v3.2.2: Python communication link active, SymPy v1.10.1.

Aproximación Taylor : 351

Valor exacto : 351

Error relativo % : 0%

```
octave:2> pregunta2
```